

MID

StatsBomb

Data Structure: *Competition → Season → Match → Events*

Player profiles are often compared using percentile ranks on a radar chart rather than raw per-90 numbers. Why is this comparison more meaningful than comparing raw stats directly?

Raw per-90 statistics do not indicate whether a player's performance is good relative to others. Percentiles compare a player with others in the same position, making it much easier to identify strengths and weaknesses and to compare metrics measured on different scales.

Expected Threat

VAEP (*Valuing Actions by Estimating Probabilities*)

Possession chain: a sequence of events where the same team has unbroken control of the ball.

Why focussing on chain rather than position? - The threat lies in how the ball is moved, rather than where the ball is.

FINAL

Optical Tracking Data

Limitations:

- Does not account for body position. (Backtracking)
- Sometimes error prone (ID switches).
- Ball usually tracked in 2d. So some strange trajectories.
- Need to link it to event data.
- Lack of public data.

Sample Questions

What is pitch control in football analytics?

Pitch control is a model in football analytics that measures the probability that a player (or team) will gain control of the ball if it is passed directly to a specific coordinate on the pitch

What is a Voronoi diagram?

- dividing space into regions based on proximity to a set of points

Define internal load in sports science.

Internal load in sports science refers to the physiological and psychological stress or cost imposed on an athlete's body in response to a training stimulus or competition

Define external load and give two examples.

External load is defined as the physical work or activities that an athlete actually performs, answering the fundamental question

Distance or velocity tracked during a session via GPS.

The number of **jumps** (such as a daily vertical jump)

What is meant by time-to-intercept?

Time-to-intercept represents the physical amount of time a player needs to run and reach a specific coordinate along the path of the ball's trajectory

What does the parameter λ represent in pass probability modeling?

the parameter λ represents the control rate. It is the constant rate at which a player secures possession of the ball once they are in close proximity to it

State one limitation of tracking data.

does not account for a player's body position,

What is sports science?

Sports science is the study of human physiology, psychology, anatomy, biomechanics, biochemistry, and biokinetics to understand how the human body works during exercise

2. Analytical Questions**1. Explain why event data alone is insufficient for analyzing off-ball player behavior.**

Event data only captures actions directly involving the player on the ball, such as passes, tackles, or shots. Because players spend the vast majority of a match without possession, event data completely misses their runs, positioning, and spatial coverage. Consequently, it cannot quantify how off-ball movements pull defenders out of position, create passing options, or control space

2. Explain the concept of passing triangles and their importance in football. (Read suggested book chapter)

Passing triangles are geometric networks formed when players space themselves to provide nearby passing options in all directions. They are crucial because they solve complex connecting problems, let players act as rapid ball-distribution junctions, and mathematically ensure the pitch is divided into well-spaced zones

3. Why are defenders advised not to stay on Voronoi boundaries?

Standing on a Voronoi boundary line makes a defender equally close to two attacking opponents. If they step out to press the ball carrier, the ball can easily be passed to the other free attacker; if they back off to cover the receiver, the ball carrier is free to drive forward. This traps the defender in a tactical "no-man's land" where they cannot successfully stop either player

4. Compare the following approaches:

- **Time-to-intercept model**
- **Machine learning model**

Focus on interpretability, flexibility, and realism.

Time-to-Intercept Model:

- **Realism & Flexibility:** Low. It is highly rigid and deterministic, calculating only who gets to the ball first without providing actual probabilities.
- **Interpretability:** Extremely high, as it is grounded strictly in physical equations of motion.

Machine Learning Model:

- **Realism & Flexibility:** High flexibility to fit complex patterns, but low physical realism (it can make predictions that violate basic laws of physics).
- **Interpretability:** Low. It is an opaque "black box" that is difficult for coaches to interpret or use for tactical training

5. Explain the roles of:

- **Time-to-intercept**
- **Time-to-control**

Why are both necessary in pass probability modeling?

In pass probability modeling, a successful pass requires two distinct steps, each modeled by a different physical-statistical parameter¹:

- **Time-to-Intercept:** This represents the probability that a player can physically run and reach the ball's path before the ball passes them¹. It is modeled using a Logistic CDF based on "spare time" (the difference between the ball's flight time and the player's run time)².
- **Time-to-Control:** This represents the probability that a player, having reached the ball, stays near it long enough to cleanly trap and secure possession¹³. It is modeled using an Exponential CDF based on a constant control rate (λ)³⁴.

Why Both Are Necessary

Reaching the ball does not guarantee controlling it³. A player—especially a challenging defender—might be fast enough to physically reach the ball's trajectory (intercepting it) but fail to secure possession, causing the ball to deflect or roll away³.

By combining both, the model realistically calculates the "horse race" for the ball⁵. It ensures that a pass is only considered successful if a teammate has the physical capability to reach the target zone first *and* is given the uncontested window of time necessary to bring the ball under control

6. Explain how simple local rules lead to complex global behavior in football movement.

Just like schooling fish, football players do not need a central coordinator to align their movements on the pitch. By simply following instinctive local rules—such as adjusting running speed, exploiting open space, and responding only to nearby teammates—highly complex global behaviors like a synchronized offside trap or a fluid, triangular passing network naturally emerge

Sardines creating living tornado around Divers

7. Why does the pass probability model ignore the Magnus force? What are the implications?

The model ignores the Magnus force to simplify ball aerodynamics, allowing the trajectory equations to assume the ball travels in a straight line rather than curving.

The main implication is a systematic calibration gap of about 11 percentage points, making the model overly pessimistic because real-world curving passes are actually much safer than the straight-line model predicts

8. Why is pitch control modeled as a probability surface instead of discrete regions?

Modeling pitch control as a continuous probability surface avoids the rigid, binary "cliff edges" characteristic of discrete Voronoi regions. This smooth mathematical design ensures that adjacent on-pitch situations receive similar probabilities, realistically reflecting spatial and physical uncertainty

9. Explain advantages and disadvantages of different external and internal load frameworks.

1. Internal Load Frameworks

- **Objective Internal Load** (e.g., Heart Rate Variability, Blood Lactate)
 - **Advantages:**
 1. Directly measures an athlete's physiological **adaptation to fitness** over time.
 2. Provides unbiased, clinical biological markers of strain.
 - **Disadvantages:**
 1. Collecting these metrics consistently can be highly **difficult and complex**.
 2. Certain methods can be physically **invasive** (e.g., taking blood lactate pricks).

External Load (e.g., GPS, Accelerometers)

- **Advantages:**
 1. Offers **great, precise measures** of actual physical work done (distances, speeds, durations).
 2. Allows for real-time monitoring to prevent immediate overtraining or injury.
- **Disadvantages:**
 1. Physically **invasive** as it requires players to wear specialized vests or sensors.
 2. High technological and financial barrier to access and process the data.

10. Is being in underload condition helpful for the players' fitness? Explain.

No, being in an underload condition is not helpful for a player's fitness.

According to the fitness-fatigue framework, underloading leads to highly negative physical adaptations, including muscle atrophy, cartilage degeneration, decreased tendon stiffness, and adipose tissue deposition, which actively degrade athletic performance.

11. What is physiological and mechanical reserve? How they get depleted? How to monitor and recover these reserves?

Physiological reserve is an athlete's cardiovascular and metabolic capacity, which is depleted by aerobic/anaerobic demands, monitored via heart rate (HR), lactate, and HRV, and recovered through rest, sleep, and nutrition.

Mechanical reserve is musculo-skeletal integrity, which is depleted by physical impacts, sprints, and jumps, monitored via GPS accelerometer loads, and recovered through structured rest days.

Failing to manage these loads leads to overload—resulting in overtraining syndrome for physiological reserve and stress fractures or muscle tears for mechanical reserve.

12. Compare modern tracking technologies? (Computer vision & optical tracking, AI-powered video analysis, wearable + optical fusion)

Here is a shorter, direct comparison of the primary tracking systems:

- **Optical Tracking (Camera-Based):** Uses stadium cameras to capture 2D coordinates and movement vectors (speed and direction) of the ball and all players on the pitch without requiring wearable sensors.
- **Wearable Tracking (GPS Vests):** Requires players to wear specialized vests or sensors to track 2D position and physiological vital data (such as heart rate). While highly detailed, it only collects data for your own team's players.

- **Integrated Tracking (Fusion):** Combines optical coordinates with wearable fitness and training metrics to give analysts a unified view of both spatial positioning and physical exertion

13. What are the forces that control ball trajectory? Why pass probability model only addresses pass that travel in straight line.

In real life, a ball's trajectory is controlled by **gravity, drag, and the Magnus (spin) force**. The pass probability model only addresses straight-line passes because it **drops the Magnus force to simplify the mathematics**, assuming the ball does not curve

3. Numerical Problems

1. A ball travels 30 meters at a speed of 10 m/s.

- Calculate the arrival time.
- Explain how this value is used in pitch control modeling.

2. A pass travels 21 meters with a speed of 15 m/s.

- Compute the ball arrival time.
- Player X reaches in 1.2 s and Player Y in 1.5 s.
- Who has a higher probability of intercepting the ball? Explain.

3. A player stays near the ball for:

- 0.1 seconds
- 0.5 seconds

Which case has a higher probability of control? Explain using intuition from the exponential model.

4. Consider a pitch scenario:

- Attacker A is 5 m away from a location.
- Defender B is 7 m away but moving faster.
- Who is more likely to control the ball?
- Explain using time-to-intercept and probability concepts.

4. Code Interpretation

```
def arrival_time ( distance , speed ) :  
    return distance / speed  
t_ball = arrival_time (20 , 10)  
t_player = arrival_time (20 , 8)  
if t_player < t_ball :  
    print ( " Player controls ball " )  
else :  
    print ( " Ball arrives first " )
```

1. What does this code compute?
2. What assumption does it make about player movement?
3. Why is this model unrealistic in real football analytics?

1. What it computes: It calculates a binary, deterministic "horse race" between the ball and a player over 20 meters using simple travel time (distance divided by speed).
2. Player movement assumption: It assumes constant velocity, meaning the player instantly reaches maximum speed (8 m/s) with zero acceleration time and zero reaction delay.
3. Why it is unrealistic: Real football analytics requires a reaction time delay (~0.7 seconds) and physically constrained acceleration limits. Furthermore, real play is highly unpredictable; modern models use smooth probability curves (Logistic and Exponential CDFs) rather than binary, rigid outcomes

5. Higher Thinking Ability

1. Explain why pitch control models are probabilistic rather than deterministic.

Pitch control models are modeled probabilistically because real-world player movement is subject to constant human and physical noise, including reaction delays, fatigue, and footing. While deterministic approaches are too rigid to capture this spatial uncertainty, a probabilistic model outputs a smooth value between 0 and 1. This design ensures that adjacent pitch situations receive similar probabilities, avoiding unrealistic, binary "cliff edges" on the field

2. How can pass probability models be used for:

- Tactical analysis
- Player evaluation

- Tactical Analysis: Heatmaps evaluate spatial control (e.g., defending set-pieces), and simulated pass variations identify mathematically optimal passing lanes.
- Player Evaluation: Expected vs. actual reception ratios measure role-based efficiency, while pass value isolates tactical decision quality from physical execution

3. Explain the concept of uncertainty in interception modeling and its importance.

Interception uncertainty represents human and physical noise—like reaction delay, footing, and fatigue—that prevents deterministic player movement

Instead of binary win-or-lose limits, it is modeled using a Logistic CDF scaled by $\sigma = 0.45s$ to give players a realistic probability of reaching the ball

This is crucial because it eliminates unrealistic "cliff edges" on the pitch, creating a continuous and mathematically stable pitch control surface

4. In a pitch control map, a defender is positioned far from his natural position (e.g., left-back appearing on the right side).

- Why might this happen?
- Explain using the concept of “defense in attack”.

Counter Attack stop korte oi places gulo block kora

5. Given a pitch visualization:



- Is it a good offside trap? Explain your answer.

Bad offside trap

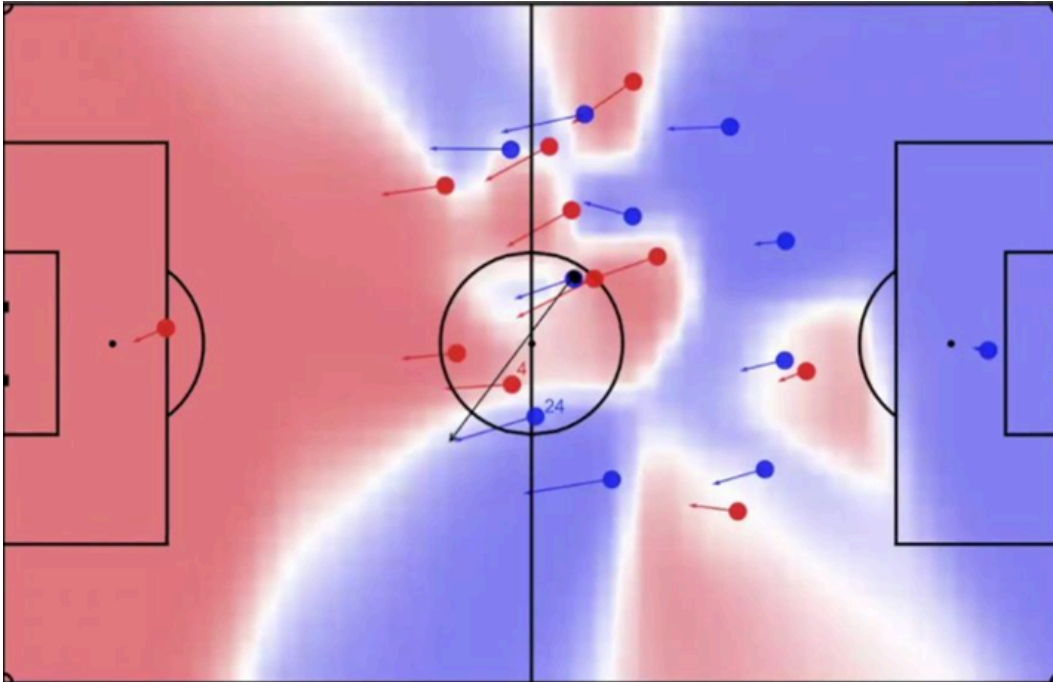
→ Not in the same line and line sight not correct

-----> responding only to nearby teammates

- How can defenders maintain offside trap?

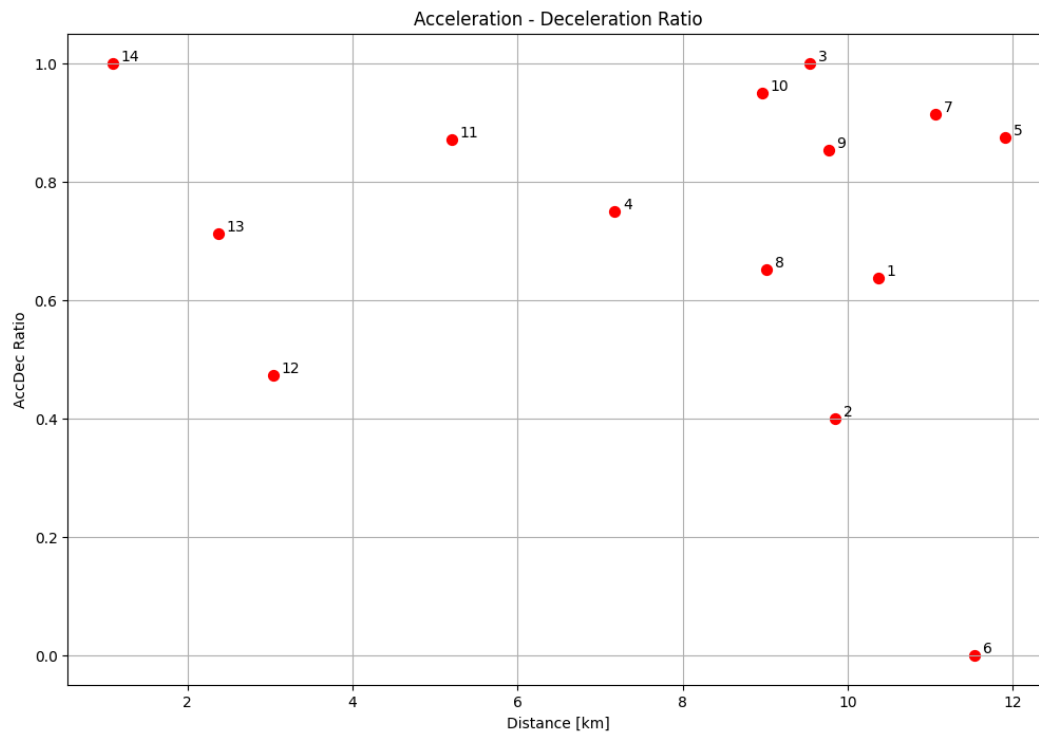
Local rules solve global complex problems

6. Comment on the quality of the pass in the picture based on pitch control model.

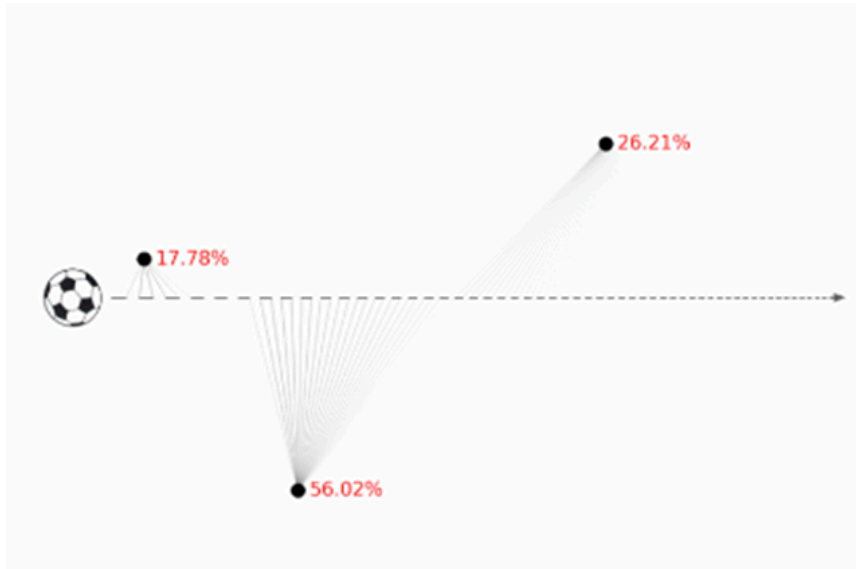


Open Space a pass

7. What can we say about the players' fitness from the plot?



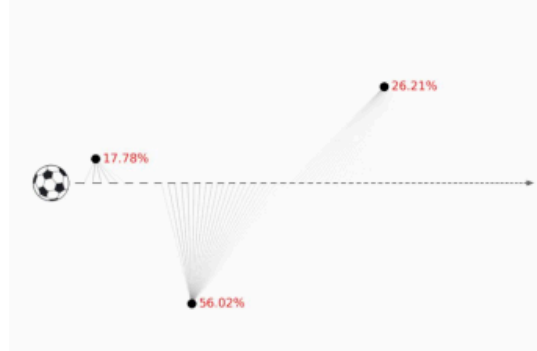
8. What can you say about the probability of the players to receive the ball?



Worked Example (Paper Figure 3)

Three players along a ball's trajectory:

Player	Position	P(receive)	Why?
1	Close to passer	17.8%	Ball still moves too fast here
2	Middle	56.0%	Best position , multiple intercept lines
3	Far down the path	26.2%	Earlier players already won the ball



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Intercept Lines + Distance

9. How can you combine Expected Possession Value (EPV) and Pitch control to create a better metrics to rate a pass? (**Refer to Spring 26 final Question 5**)

10. What does the following code block do? Explain the process. How can it improve the xG modeling?

```
def players_in_triangle(test_shot, track_df):  
    #get id of the shot to search for tracking data using this index  
    test_shot_id = test_shot["id"]  
    #get all opposition's player location  
    player_position = track_df.loc[track_df["id"] == test_shot_id].loc[track_df["teammate"] ==  
False][["x", "y"]]  
    #checking if point inside a triangle  
    x1 = 105  
    y1 = 34 - 7.32/2  
    x2 = 105  
    y2 = 34 + 7.32/2  
    x3 = test_shot["x"]  
    y3 = test_shot["y"]  
    xp = player_position["x"]  
    yp = player_position["y"]  
    c1 = (x2-x1)*(yp-y1)-(y2-y1)*(xp-x1)  
    c2 = (x3-x2)*(yp-y2)-(y3-y2)*(xp-x2)  
    c3 = (x1-x3)*(yp-y3)-(y1-y3)*(xp-x3)  
    #get number of players inside a triangle  
    return len(player_position.loc[((c1<0) & (c2<0) & (c3<0)) | ((c1>0) & (c2>0) & (c3>0))])
```

Passing triangles are formed when players position themselves in triangular networks, ensuring the ball-carrier always has nearby passing options in multiple directions to circulate the ball quickly and open up space.

This improves xG modeling because incorporating team passing rates, possession chains, and spatial structures into traditional shot-only models provides crucial build-up context, predicting future shot quality and actual goal-scoring rates far more accurately than shot locations alone

(a) [2 marks] Define sports science.

(b) [4 marks] A sports scientist uses a performance evaluation form completed by a coach after a training session.

Classify this data collection method along two dimensions: (i) Internal vs. External load, and (ii) Subjective vs.

Objective measurement. Describe the advantages and disadvantages of this framework.

(c) [2 marks] What can acceleration-deceleration data tell us about a player's fitness?

1. Total Number of Goals in the Match

A score of 2 – 2 means that exactly $k = 4$ goals were scored in total across the 90 minutes.

From the question:

- Number of independent trials (minutes): $n = 90$
- Average number of goals per match: $\lambda = 2.5$
- Probability of a goal being scored in any given minute:

$$p = \frac{2.5}{90} = \frac{1}{36} \approx 0.02778$$

Using the binomial probability formula given in the question:

$$P(K = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

- **Combinations:**

$$\binom{90}{4} = \frac{90 \times 89 \times 88 \times 87}{4 \times 3 \times 2 \times 1} = 2,555,190$$

- **Probabilities:**

$$p^4 = \left(\frac{1}{36}\right)^4 \approx 5.9527 \times 10^{-7}$$

$$(1 - p)^{86} = \left(1 - \frac{1}{36}\right)^{86} = \left(\frac{35}{36}\right)^{86} \approx 0.08906$$

- **Probability of exactly 4 total goals:**

$$P(K = 4) \approx 2,555,190 \times (5.9527 \times 10^{-7}) \times 0.08906 \approx 0.1355 \quad (13.55\%)$$

2. Goal Distribution Between the Two Teams

The problem states: "Assume each goal is equally likely to be scored by either team, independently."

Given that 4 goals are scored, the number of goals scored by Team A follows a binomial distribution with parameters $N = 4$ and probability $p_{\text{team}} = 0.5$:

$$P(\text{Team A scores 2} \mid 4 \text{ total goals}) = \binom{4}{2} (0.5)^2 (0.5)^2 = 6 \times \left(\frac{1}{2}\right)^4 = \frac{6}{16} = 0.375$$

3. Final Probability of a 2–2 Draw

The total probability of the match ending 2 – 2 is:

$$P(2 - 2) = P(K = 4) \times P(\text{Team A scores 2} \mid K = 4)$$

$$P(2 - 2) \approx 0.135508 \times 0.375 \approx \mathbf{0.0508}$$

Answer:

The probability that a match ends 2 – 2 is approximately 0.0508 (or 5.08%).

(Note: If approximated via the Poisson distribution $\frac{\lambda^k e^{-\lambda}}{k!}$, $P(K = 4) = \frac{2.5^4 e^{-2.5}}{24} \approx 0.1336$, yielding $P(2 - 2) \approx 0.1336 \times 0.375 \approx \mathbf{0.0501}$ or **5.01%**).

Coding Files

Pitch Control